

ABHI PATEL

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SUMMARY

Data Scientist specializing in NLP, predictive modelling, and ML deployment. Three years of project and hands-on experience delivering end-to-end pipelines in Python, with certifications in AWS, Google GenAI, and Anthropic Agentic AI. Currently completing a Graduate Certificate in AI/ML at Lambton College. Comfortable across the full pipeline from data wrangling and feature engineering to model evaluation, API deployment, and communicating findings to non-technical stakeholders.

TECHNICAL SKILLS

Languages & Data: Python, SQL, Pandas, NumPy, Scikit-learn

Machine Learning: Regression, classification, clustering, NLP, feature engineering, cross-validation, model deployment, ETL pipelines

Deep Learning & Computer Vision: TensorFlow, CNNs, VGG16, transfer learning, AutoKeras

GenAI & LLMs: Hugging Face Transformers, BERT, Word2Vec, embeddings, vector databases, LangChain, prompt engineering

MLOps & Cloud: AWS (EC2, S3, IAM, VPC), Azure, Docker, Git, Flask, REST APIs, MLflow

Visualization & Reporting: Matplotlib, Seaborn, Jupyter Notebooks, Excel, PowerPoint

AI Coding Tools: GitHub Copilot, Claude

PROFESSIONAL EXPERIENCE

Web Application Developer (Volunteer)

Hari Prabodham Mandir, Toronto, ON | Jan 2024 - Present

- Engineered a scalable data-management and registration system using Python, automating workflows for 100+ community members.
- Consolidated disparate legacy spreadsheets into a centralized structured database, eliminating manual data entry errors and boosting data integrity.
- Collaborated closely with non-technical stakeholders through iterative feedback sessions to translate ambiguous community requirements into practical software solutions.

PROJECTS

Smart Auto: Predictive Pricing Model

Python, Pandas, NumPy, Scikit-learn, SQL, Flask

- Developed a regression model on a dataset of 23,000+ vehicle records to predict resale pricing, achieving an R-squared score of 0.89 on test data.
- Optimized data ingestion pipelines by writing complex SQL queries and leveraging Pandas to aggregate, clean, and feature-engineer dataset, handling missing values and outliers to ensure data integrity.
- Deployed the model behind a Flask REST API, enabling non-technical users to get real-time price predictions through a clean web interface.
- Benchmarked linear regression, random forest, and gradient boosting; documented accuracy trade-offs and justified the final model selection with evidence.

Exam Question Generator

Python, BERT, Word2Vec, Hugging Face Transformers, Scikit-learn

- Designed an end-to-end NLP pipeline to group questions by semantic meaning using BERT embeddings and clustering algorithms, reducing manual categorization time by an estimated 70%.
- Compared BERT contextual embeddings against Word2Vec static embeddings across multiple clustering configurations; documented which approach produced more coherent clusters and why.
- Preprocessed and vectorized a corpus of unstructured text, then built a semantic retrieval layer on top of the embeddings, gaining hands-on experience with vector database mechanics.

- Implemented multi-step reasoning logic to automate exam generation from clustered results, producing a small-scale agentic workflow for automated content creation.

Medical Image Analysis: Model Evaluation

Python, TensorFlow, VGG16, AutoKeras, Matplotlib

- Fine-tuned VGG16 and used AutoKeras NAS for two medical image classification tasks, reaching 86-92% accuracy across both datasets.
- Ran systematic experiments varying preprocessing, data augmentation, and architecture; documented results in comparison tables and charts for a non-technical audience.
- Used AutoKeras to automate architecture search across hyperparameter combinations, cutting experiment scoping time by approximately 40% compared to manual tuning.

EDUCATION

Graduate Certificate, Artificial Intelligence & Machine Learning

Lambton College, Toronto, ON | Jan 2024 - Aug 2025

Coursework: Supervised and unsupervised learning, deep learning, NLP, data engineering, cloud and big data, model deployment

Bachelor of Computer Applications (BCA)

Shri B. J. Patel College, Gujarat, India | Jan 2019 - Apr 2022

Coursework: Programming (C, C++, Java, Python), Data Structures, DBMS, Operating Systems, Computer Networks, Web Development, Software Engineering, Mathematics for Computing

CERTIFICATIONS

AWS Cloud Architecture - Amazon Web Services | Apr 2026

Cloud infrastructure, VPC design, IAM, EC2, S3, load balancing, and scalable architecture patterns

Generative AI - Google | Apr 2026

LLM fundamentals, prompt engineering, responsible AI, and practical GenAI workflows

Claude & Agentic AI – Anthropic | Dec 2025

Agentic workflows, prompt engineering, tool use orchestration, and safe LLM deployment practices